Independent Bookstore System with Reader Club

FUNCTIONAL REQUIREMENTS

What the system should DO

GUEST USERS

FR-01	Register new account with email verification
FR-02	Browse books by title, author, or genre
FR-03	Search books with real-time filtering
FR-04	View book details (price, description, ratings)

REGISTERED USERS

FR-05	Secure login/logout with session management
FR-06	Add books to persistent shopping cart
FR-07	Manage cart (update quantities, remove items)
FR-08	Place orders with shipping address input
FR-09	View order history and status tracking
FR-10	Write reviews (1-5 stars) for purchased books only

READER CLUB FEATURES

FR-11	Create and join book clubs (public/private)
FR-12	Collaborative reading schedule with milestone tracking

NON-FUNCTIONAL REQUIREMENTS

How the system should BEHAVE

PERFORMANCE

NFR-01	Pages load within 3 seconds under normal load
NFR-02	System supports up to 100 concurrent users without degradation
NFR-03	Search results return in less than 1 second
NFR-04	Database queries execute in under 200ms

SECURITY

NFR-05	Passwords hashed using bcrypt (cost factor = 10)
NFR-06	Prepared statements for all database queries (SQL injection prevention)
NFR-07	Session tokens expire after 24 hours of inactivity
NFR-08	Role-based access control enforced at API level

USABILITY

NFR-09	Fully responsive design (mobile, tablet, desktop)
NFR-10	Clear error messages with suggested actions

RELIABILITY & SCALABILITY

NFR-11	99.5% uptime during business hours

FR-13	Vote for next book using weighted polling system
FR-14	Member spotlight algorithm (Reader of the Month)
FR-15	Reading challenge manager (e.g., 52 books/year)

ADMINISTRATORS

FR-16	Admin dashboard with role-based access control
FR-17	CRUD operations for book inventory management
FR-18	View and update all customer orders
FR-19	Manage user accounts (activate/deactivate, promote)
FR-20	Generate basic reports (top reviewed books, sales)

SYSTEM BUSINESS RULES

FR-21	Stock validation before order confirmation
FR-22	Automatic average rating calculation
FR-23	Purchase verification for review eligibility
FR-24	Multi-vendor cart splitter for orders from multiple stores
FR-25	Cross-store inventory sync engine (real-time)

NFR-12	Transaction rollback on any stock validation failure
NFR-13	Modular architecture allowing horizontal scaling

DESIGN & IMPLEMENTATION CONSTRAINTS

C-01	Backend: PHP 8+ with secure coding practices
C-02	Database: MySQL 8+ with optimized queries
C-03	Frontend: HTML5/CSS3/JavaScript (vanilla, no heavy frameworks)
C-04	Local development: XAMPP/WAMP
C-05	Production-ready code structure with easy cloud migration